

JIAJIE (JERRY) LUO

Knowledge Lab ◇ University of Chicago

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Last update: September 29, 2026

EMPLOYMENT

Postdoctoral Scholar

October 2024 – Present

Knowledge Lab

University of Chicago

Mentor: Professor James Evans

Ph.D. Research Intern

June 2022 – September 2022

Mathematics, Statistics, and Data Science

Pacific Northwest National Laboratory

Mentors: Dr. Tegan Emerson; Dr. Gregory Henselman-Petrusek Roek

EDUCATION

University of California, Los Angeles

September 2019 – June 2024

Ph.D. in Mathematics.

Thesis: Topics in Persistent Homology and Complex Social Systems

Advisor: Professor Mason Porter

University of California, Santa Barbara

September 2017 – June 2019

M.A. in Mathematics.

Thesis: On Abstract Witt Rings and Quadratic Extensions

Advisor: Professor Bill Jacob

University of California, Santa Barbara

September 2014 – June 2017

College of Creative Studies

B.S. in Mathematics, *Highest Honors*

Faculty Advisor: Professor Jeffrey Stopple

RESEARCH INTERESTS

Theory and Applications of Topological Data Analysis, Persistent Homology, Models of Opinion Dynamics, Network Science, Complex Systems, Computational Social Science, Data Science

PREPRINTS & PUBLICATIONS

(*Equal Contribution)

(†Mentored Student)

1. S. Tymochko, G. Grindstaff*, A. Hickok*, **J. Luo***, M. A. Porter, “Using Persistent Homology to Analyze Access to Heterogeneous-Quality Resources and Heterogeneous-Severity Nuisances,” *arXiv:2609.34090* (Preprint), 2026.

2. **J. Luo**, J. A. Evans, “Filling Holes in Science Draws Collective Attention, But Most Higher-Order Holes Remain Unexplored,” *arXiv:2608.28822* (Preprint), 2026.
3. L. Thompsky[†], Y. Wu[†], M. A. Porter, **J. Luo**, “A Bounded-Confidence Model of Opinion Dynamics with Adaptive Interaction Probabilities,” *arXiv:2605.20418* (Preprint), 2026.
4. **J. Luo**, G. Henselman-Petrusek, “Interval Decomposition of Infinite Persistence Modules over a Principal Ideal Domain,” *Proceedings of the American Mathematical Society*, Vol. 154, No. 11, pp. 4931–4945, 2026.
5. G. J. Li, **J. Luo**, W. Chu, “Bounded-Confidence Models of Multi-Dimensional Opinions with Topic-Weighted Discordance,” *SIAM Journal on Applied Dynamical Systems*, Vol. 25, No. 1, pp. 1–41, 2026.
6. **J. Luo**, G. Henselman-Petrusek, “Interval Decomposition for Persistence Modules Over a Principal Ideal Domain,” *Foundations of Computational Mathematics*, Vol. 26, No. 4, pp. 2571–2623, 2026.
7. G. J. Li*, **J. Luo***, M. A. Porter, “Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds,” *SIAM Journal on Applied Dynamical Systems*, Vol. 24, No. 2, pp. 994–1041, 2025.
8. A. Hickok*, B. Jarman*, M. C. Johnson*, **J. Luo***, M. A. Porter, “Persistent Homology for Resource Coverage: A Case Study of Access to Polling Sites,” *SIAM Review*, Vol. 66, No. 3, pp. 481–500, 2024.
9. V. Chayes, K. Miller, R. Bhalerao, **J. Luo**, W. Zhu, A. Bertozzi, W. Liao, S. Osher, “Pre-Processing and Classification of Hyperspectral Imagery Via Selective inpainting,” *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, New Orleans, LA, USA, 2017.

WORKS IN PREPARATION

(*Equal Contribution)

([†]Mentored Student)

1. Y. Yang*, **J. Luo***, M. A. Porter, J. H. Tien, “A Hegselmann–Krause Opinion Model Coupled with Susceptible–Infected–Recovered Disease Dynamics.”
2. W. Flowers[†], **J. Luo**, “Bounded-Confidence Models of Opinion Dynamics with Adaptive Compromise.”
3. Z. Zhao[†], **J. Luo**, “A Hegselmann–Krause Model of Opinion Dynamics with Network Rewiring.”
4. **J. Luo**, J. A. Evans, R. Funk, L. Ziegelmeier, “The Process of Hole Formation in Science.”
5. **J. Luo**, J. A. Evans, “A Mapper Graph of Scientific Knowledge.”

EXPOSITORY ARTICLES

G. J. Li, **J. Luo**, K. Peng, and M. A. Porter, “Using Mathematics to Study How People Influence Each Other’s Opinions,” *Frontiers for Young Minds*, Vol. 12, Article 1253153, 2024.

AWARDS, HONORS & FELLOWSHIPS

Pacific Journal of Mathematics Dissertation Prize (UCLA)	2024
ModELing and uNdersTanding human behaviOR (MENTOR) Fellowship (UCLA)	2021–2022
College of Creative Studies Commencement Speaker (UCSB)	2017
Adil Yaqub is my Hero Scholarship (UCSB)	2016

TALKS & PRESENTATIONS

Joint Mathematics Meetings (JMM 2027) January 2027
AMS Special Session: Mathematics of Complex Data Models in Scientific Workflows
 A Persistent-Homology Approach for Identifying Knowledge Gaps (Invited Talk)

SIAM Conference on Mathematics of Data Science (MDS26) November 2026
Complex Social Systems (Minisymposium)
 A Persistent-Homology Approach for Identifying Knowledge Gaps (Invited Talk)

Applied Algebraic Topology Research Network (AATRN) Poster Session September 2026
 Interval Decomposition of Persistence Modules over a Principal Ideal Domain (Virtual Poster Session)

Union College Mathematics Conference September 2026
 Interval Decomposition of Persistence Modules over a Principal Ideal Domain (Contributed Talk)

SIAM Annual Meeting (AN26) July 2026
Advances in Modeling and Inference for Dynamics on Networks (Minisymposium)
 Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds (Invited Talk)

LANS Seminar, Argonne National Laboratory April 2026
 Generalizations and Extensions of Bounded-Confidence Models (Invited Talk)

IMSI Workshop: The Geometric Realization of AATRN August 2025
 Interval Decomposition of Persistence Modules over a Principal Ideal Domain (Poster + Lightning Talk)

IMSI Workshop: Emergent Behavior in Complex Systems of Interacting Agents March 2025
 Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds (Poster Session)

Southern California Applied Mathematics Symposium (SOCAMS) April 2024
 Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds (Contributed Talk)

Graduate Student Topology and Geometry Conference (GSTGC2024) April 2024
 Interval Decomposition of Persistence Modules over a Principal Ideal Domain (Poster Session)

Joint Mathematics Meetings (JMM 2024) January 2024
AMS Special Session: Complex Social Systems
 Persistent Homology for Assessing Facility Placement (Invited Talk)

2023 Algorithms for Threat Detection PI Workshop (ATD2023) October 2023
 Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds (Contributed Talk)

Computation Persistence Workshop (ComPer23) September 2023
 Interval Decomposition for Persistence Modules of Free Abelian Groups (Contributed Talk)

SIAM Conference on Applications of Dynamical Systems (DS23) May 2023
 Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds (Contributed Talk)

Southern California Applied Mathematics Symposium (SOCAMS) April 2023
 Persistent Homology for Resource Coverage: A Case Study of Access to Polling Sites (Contributed Talk)

SIAM Workshop on Network Science (NS22) September 2022
 Bounded-Confidence Models with Adaptive Confidence Bounds (Contributed Talk)

Virtual Research Symposium, Pacific Northwest National Laboratory. August 2022
 Topological Data Analysis and Machine Learning (Contributed Talk)

TEACHING EXPERIENCE

As Instructor of Record (UCLA)
 Math 110A: Abstract Algebra

Winter 2024

Math 115A: Linear Algebra (proof-based)	Winter 2023
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As Graduate Teaching Assistant (UCLA)

Math 31AL: Differential and Integral Calculus Laboratory	Winter 2021
Math 115A: Linear Algebra (proof-based)	Fall 2020, Spring 2021
Math 31B: Integration and Infinite Series	Spring 2020
Math 33A: Linear Algebra and Application	Winter 2020, Fall 2020, Spring 2021
Math 3B: Calculus for Life Sciences II	Fall 2019, Winter 2021

As Graduate Teaching Assistant (UCSB)

Math 117: Methods of Analysis	Spring 2019
Math 108A: Introduction to Linear algebra (proof-based)	Winter 2019
Math 4A: Linear Algebra and Applications	Fall 2018
MATH 100B: Mathematics for Elementary Teaching II	Summer 2018
Math 34B: Calculus for Social Sciences II	Winter 2018, Spring 2018
Math 34A: Calculus for Social Sciences I	Fall 2017

STUDENT MENTORING

Research Mentoring (Students from UChicago):

Ziqian Zhao — A Bounded-Confidence Model with Network Rewiring	Fall 2025 – Present
Joy Yang — The Effects of Topic Shifts on Researchers	Fall 2025 – Present
Xiang (Sherry) Ding — The Effects of Topic Shifts on Researchers	Fall 2025 – Present
William Flowers — Bounded-Confidence Models of Opinion Dynamics	Fall 2024 – Present

Research Mentoring (Students from UCLA):

Yuxuan Wu (with Mason A. Porter) — A Bounded-Confidence Model with Adaptive Edge Weights	Summer 2024 – Spring 2026
Leila Thompsky (with Mason A. Porter) — A Bounded-Confidence Model with Adaptive Edge Weights	Fall 2023 – Spring 2026
Amos Ancell — Persistent Homology for Resource Coverage	Fall 2023 – Spring 2024
Ruyi Lu — Bounded-Confidence Models on Random Configuration Models	Winter 2023 – Fall 2023
Xinyue (Serena) Li — Persistent Homology for Resource Coverage	Winter 2023 – Spring 2023
Xiaohe (Haley) Zhang (with Mason A. Porter) — Bounded-Confidence Models with Repulsion	Winter 2022 – Spring 2022

Directed Reading Program:

Member of DRP Committee	Fall 2021 – Spring 2024
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Students:

Yuxuan (Yolanda) Wu — Models of Opinion Dynamics	Spring 2024
Leila Thompsky — Complex Social Systems	Fall 2023
Amos Ancell — Applied Topology, Persistent Homology	Winter 2023 – Spring 2023
Xinyue (Serena) Li — Applied Topology, Persistent Homology	Fall 2022 – Winter 2023
Ruyi Lu — Opinion Dynamics on Networks	Fall 2022 – Winter 2023
Haoyang Lyu — Applied Topology, Persistent Homology	Winter 2022 – Spring 2022
Chenxin (Amy) Shen — Applied Topology, Persistent Homology	Fall 2021 – Spring 2022
Xiaohe (Haley) Zhang — Opinion Dynamics on Networks	Fall 2021 – Winter 2022
Tanishq Bhatia — Topics in Persistent Homology	Winter 2021 – Spring 2021

Other Mentoring:

Mentor for UCLA Applied Mathematics REU	Summer 2021
ATD Traffic Challenge, joint with Mason A. Porter	

Students: Matthew Hudes (Tufts University), Naji Sarsam (UCLA),
Chenxin (Amy) Shen (UCLA), Wenwen Tang (USC)

REFERENCES

Professor Mason A. Porter Nordita, Stockholm University	mason.porter@su.se
Professor James A. Evans University of Chicago	jevans@uchicago.edu
Dr. Gregory Henselman-Petrusek Roek Pacific Northwest National Laboratory	gregory.roek@pnnl.gov
Professor Marcus Roper (<i>teaching</i>) University of California, Los Angeles	mroper@math.ucla.edu

MISCELLANEOUS

Citizenship: United States
Programming Experience: Python, MATLAB, R, C++
Date of Birth: 3 June 1996
Languages: Chinese (Mandarin), English